

Development of a receiver-agnostic and real-time jamming and spoofing detection and mitigation system

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BIOGRAPHY (IES)

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Nabeel Ali Khan is developing digital signal processing algorithms for applications involving signal detection, instantaneous frequency estimation, jammer interference characterization and mitigation, design of time-frequency representation, positioning using signal of opportunity.

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ABSTRACT

GNSS has been providing global positioning, navigation and time (PNT) solutions for many application domains covering safety critical, liability critical and commercial applications. However, due to the relatively low-power signal around -128dBm (sensitive to jamming) and the openness of the signal structures (sensitive to spoofing), GNSS signals are vulnerable to jamming and spoofing attacks. These attacks significantly disrupt GNSS services by rendering them unavailable or generating invalid PNT solutions that might cause severe impacts on assets or users. Specifically for spoofing, with the growing capability and affordability of software-defined radio (SDR) and advanced algorithms, spoofing attacks have become more significant. This paper presents the development and demonstration of a receiver-agnostic and real-time signal cleaner system utilising innovative and efficient algorithms as well as advanced machine learning methods for jamming and spoofing detection and mitigation of GNSS signals.

Results from testing campaigns are as follows. For jamming, the probability of jamming detection of the system is >0.98 with a false alarm rate of 0.0013 and 10^{-7} for chirp and CDMA interferences, respectively, for a jammer-to-noise ratio of 5 dB and above. For the low-powered chip jammer, the system can mitigate the jammer with up to a 5 dB C/N0 improvement. For spoofing, the detection accuracy achieved with the TEXBAT dataset is $>99\%$ for 2-class spoofing classification (authentic or spoofed) and around 97% for 3-class classification (authentic, non-aligned spoofed, aligned spoofed). The false alarm rate of the spoofing detector is 0.009 for a clean-sky clean signal and 0.015 for an urban authentic signal with heavy multipath impairments. The spoofing mitigation results show a reduction in static position error by up to 70% while also improving the duration of position availability. In conclusion, we have demonstrated that the developed real-time and receiver-agnostic jamming and spoofing detection and mitigation system is effective on real-world data. The system has been validated using a professional (commercial) GNSS receiver.

INTRODUCTION

GNSS has been providing global positioning, navigation and time (PNT) solutions for many application domains covering safety critical, liability critical and commercial applications. However, due to the relatively low-power signal around -128dBm (sensitive to jamming) and the openness of the signal structures (sensitive to spoofing), GNSS signals are vulnerable to jamming and spoofing attacks [1,2]. These attacks significantly disrupt GNSS services in terms of not being available or producing invalid PNT solutions that might cause severe impacts on assets or people using the services. There are various well-known interferences and jamming attacks including broadband (such as CDMA) jamming, frequency-hopping jamming, narrow-band jammer and frequency-sweep or chirp (and pulsed chirp) jamming attacks as well as spoofing attacks ranging from a brute force spoofer transmitting GNSS-like signals at higher power, meaconing (replay), aligned to overlapping and sophisticated spoofing attacks [3]. These interferences could be non-intentional, such as frequency interference from communication signals or intentional, such as chirp jamming. Meanwhile in spoofing, the attack is most likely intentional. Currently, with the growing capability and affordability of software-defined radio (SDR) and advanced spoofing generation algorithms, spoofing attacks realisations have become more significant. Motivated by the need at providing resilient GNSS operation in vulnerable environments facing jamming and spoofing threats, there is a need to develop a receiver-agnostic and flexible signal cleaner system to increase the resilience of GNSS services.

Motivated by the need for resilient navigation in environments threatened by GNSS interference and spoofing, this paper aims to develop and demonstrate a receiver-agnostic, real-time signal cleaner system that utilises innovative and efficient algorithms for detecting, characterising, and mitigating jamming and spoofing signals. For spoofing, the mitigation focuses on overlapping (aligned) spoofing attacks on GPS L1 C/A signal. **FIGURE 1** shows the schematic view of the signal cleaner system. In this system, the developed jamming and spoofing detection and mitigation algorithms use optimised signal processing techniques and advanced machine learning algorithms. The developed system is configurable and uses commercial-off-the-shelf (COTS) components including a GNSS antenna, a high-performance computer for real-time jamming and spoofing detection and mitigation processing, a radio-frequency front-end to receive and transmit GNSS signals and a downstream commercial receiver to receive and process the mitigated/cleaned signal from the system. The system can operate in real-time up to a 20MHz sampling rate and is designed for flexibility, allowing for seamless integration between a GNSS antenna and a receiver without requiring any modifications to the receiver's existing infrastructure.

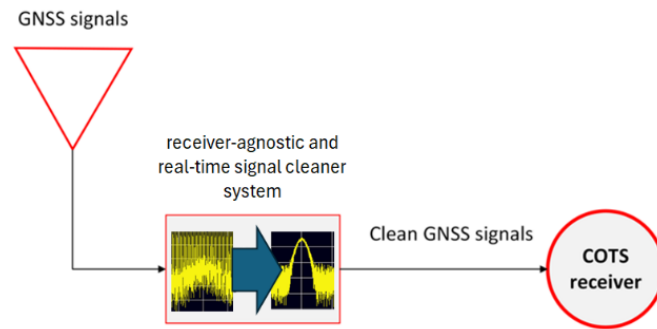


FIGURE 1: Receiver-agnostic and real-time signal cleaner system.

The main contributions of this paper are as follows:

- The development of an efficient and effective machine-learning (ML) based jamming and spoofing detection algorithm utilising advanced neural-network techniques,
- The development of jamming and spoofing mitigation algorithms using novel approaches,
- The development of a buffering system enabling real-time software implementation (with RF-input and RF-output) of the detection and mitigation algorithms,
- The implementation of a real-time jamming (including unintentional interferences) and spoofing detection and mitigation (signal spectrum cleaner) system which supports an RF-input and RF-output with up to 20MHz sampling rate by using COTS hardware.

The structure of this paper is as follows. The next two sections discuss the methods for jamming and spoofing detection and mitigation including the development of the ML methods for detection. The following section presents the development of the signal spectrum cleaner including data buffering system to support RF-to-RF Realtime processing. Real-world experiment results to test the developed system are presented and discussed afterwards. Finally, the conclusion and future work of the activity are presented.

JAMMING DETECTION AND MITIGATION

The overall jamming detection and mitigation system is composed of two modules a) interference detection module and b) interference mitigation module as illustrated in **FIGURE 2**.

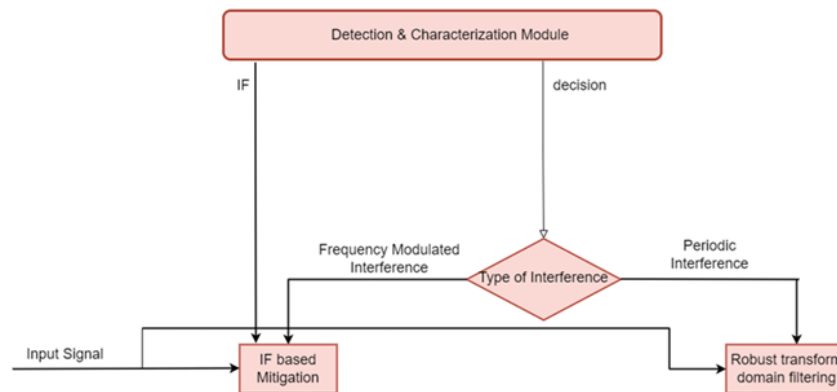


FIGURE 2: Jamming interference detection and mitigation system.

Jamming detection and characterisation

The interference detection and characterization system extract a) periodicity detection feature and b) frequency modulation features from a segment of received signal for detection and characterization of interference as illustrated in **FIGURE 3**.

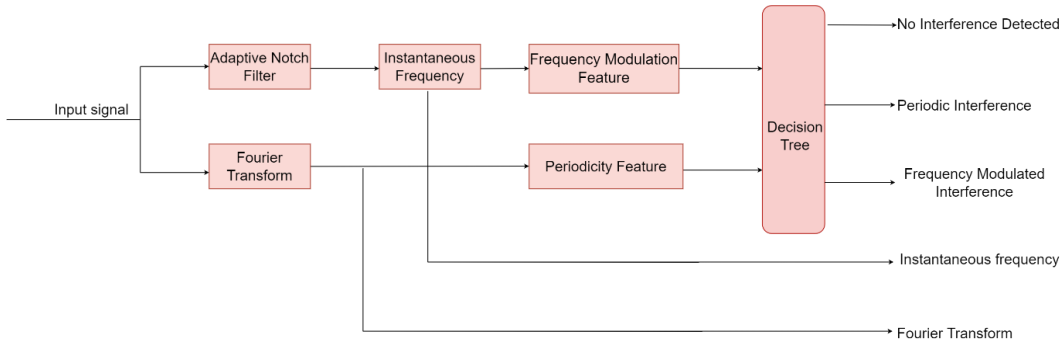


FIGURE 3: Interference characterisation module.

Periodicity Detection Feature: The periodicity check is performed to analyze whether the interfering signal is periodic that is $x[n] = x[n + K]$ or aperiodic. Note that a chirping signal can become a periodic signal if the same sequence is repeated over time. However, this is not always the case. In case of periodic interference, the interfering signal becomes sparse in the frequency domain and appears as a train of impulses as shown in FIGURE 4.

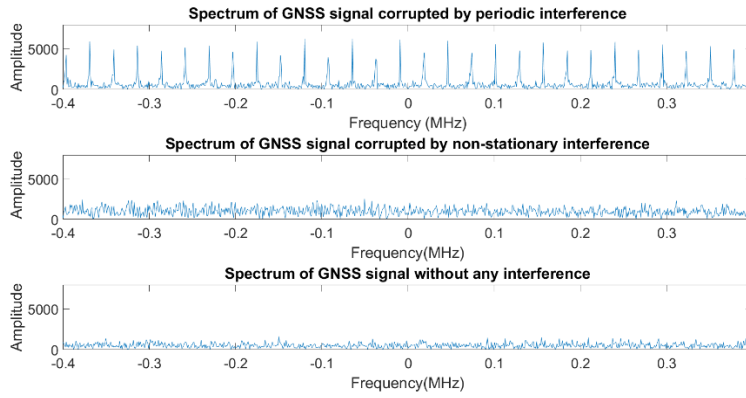


FIGURE 4: Frequency domain characterisation of periodic jammer, non-periodic jammer, and jammer free GNSS signal.

To find out whether the interfering signal is periodic or non-periodic, we first transform a given signal of N_F samples in frequency domain as $X[k] = \sum_n x[n] e^{-j2\pi nk/N_F}$. The parameter N_F is based on the periodicity interval, i.e., K of the interference signal. It should be at least 8 times the periodicity interval of the interference signal, $N_F > 8K$. We detect a M number of strong peaks from the $|X[k]|^2$ and store them in $Y[k]$. The periodicity detection feature is defined as:

$$F_1 = \frac{\text{median}(Y[k])}{\text{mean}(|X[k]|^2)} \quad (\text{eq. 1})$$

In case of periodic interference F_1 has a higher value whereas for non-periodic interference or noise, F_1 has a lower value.

Frequency Modulation Feature: Frequency modulated jammers are modelled as $J(t) = Ae^{j\int f(\tau)d\tau}$, where A is the signal amplitude and $f(t)$ is the instantaneous frequency (IF). To extract the frequency modulation feature, we first employ the adaptive notch filter (ANF) to track the IF curve. The IF information is then used to de-chirp $J(t)$ and the frequency modulation detection feature is then extracted based on its covariance matrix.

The steps to estimate the frequency modulation feature are as follows:

IF estimation: The IF is defined as derivative of the phase of the incoming signal implying it can be simply estimated as: $f_0[n] = \angle x[n]x^*[n-1]$. The ANF based IF estimation method adopts a recursive approach where a past estimate of the IF is used as a predicted value and phase difference between the current sample of signal and past sample of its bandpass filtered version are employed to correct the predicted IF [4]:

$$f_0[n] = f_0[n-1] + 2u\angle(x[n]x_r^*[n-1]e^{-jf_0[n-1]}) \quad (\text{eq. 2})$$

Where u is update rate, $x_r[n]$ is bandpass filtered signal obtained by placing pole close to $f_0[n]$ that suppresses out of the band noise:

$$x_r[n] = (1 - k_\alpha)x[n - 1] + k_\alpha x_r[n]e^{jf_0[n]} \quad (\text{eq. 3})$$

Where k_α is filter bandwidth control parameter.

Once crude IF is estimated using abovementioned method, the refined estimate is obtained by using $f_0[n]$ as the predicted value and performing correction based on the moving average of the phase differences of $x[n]$ and $x_r[n - 1]$ as discussed in [4]:

$$f[n] = f_0[n] + \frac{\alpha}{2M+1} \sum_{k=-M}^M \angle(x[n+k]x_r^*[n+k-1]e^{-jf_0[n+k]}) \quad (\text{eq. 4})$$

Feature Extraction: Once the IF is estimated, the corresponding instantaneous phase is estimated as: $\hat{\theta}[n] = \sum_{k=0}^n f[k]$. The estimated instantaneous phase is used to de-chirp the signal as $x_c[n] = x[n]e^{-j\hat{\theta}[n]}$. This de-chirped signal has now high temporal correlation as illustrated in **FIGURE 5**.

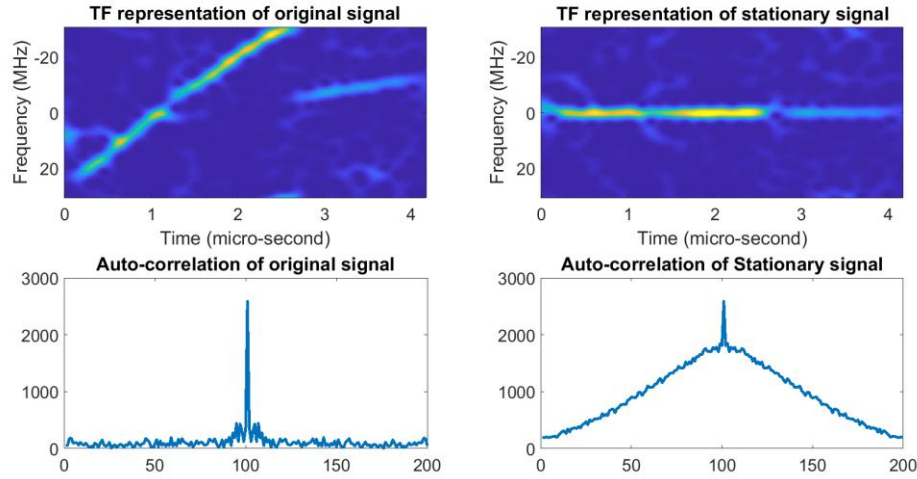


FIGURE 5: Illustration of de-chirping operation in TF domain and its impact on autocorrelation.

The covariance matrix of $x_c[n]$ is computed as: $R[n, m] = \left| \frac{1}{N} \sum_{k=0}^N x_c[n+k]x_c^*[m+k] \right|$. In case of a frequency modulated jamming signal, all the elements of $R[n, m]$ will have values significantly greater than zero whereas for GNSS signals non-diagonal elements would be close to zero. So, the ratio of sum of diagonal elements of $R[n, m]$ versus the sum of all the elements can be used as a test statistic [4] of F_2 will have a higher value for pure GNSS signal and lower value for GNSS signals corrupted by noise, as follows:

$$F_2 = \frac{\sum_n R[n, n]}{\sum_n \sum_k R[n, k]} \quad (\text{eq. 5})$$

Decision Tree based Classification: To solve a multi-class classification problem of detecting periodic and frequency modulated interferences, we train a machine learning classifier. Training dataset is generated by adding interferences to recorded GNSS signals of 100ms duration and 15MHz bandwidth. The following type interferences are used to train the classifier which are chirp type non-stationary interference and periodic interference. The decision tree is trained using the frequency modulation feature and periodicity feature for the following classes: (1) Periodic interference, (2) Frequency modulated interference and (3) Interference free GNSS signal. The decision tree thus obtained is presented in **FIGURE 6**.

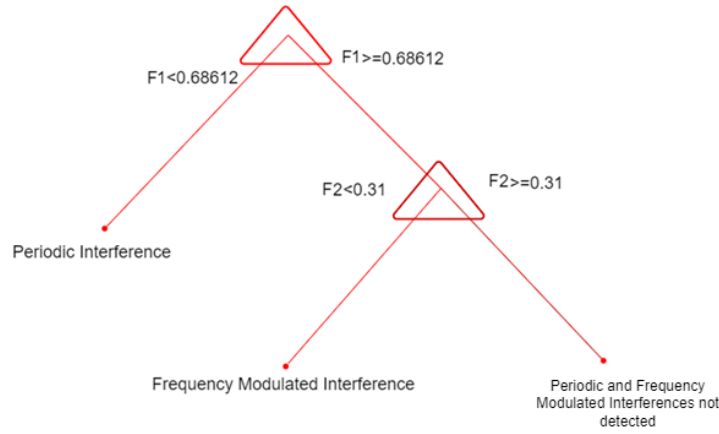


FIGURE 6: Figure 6 Decision tree illustrating characterization of jamming attack.

Jamming mitigation

The interference mitigation unit employs the outputs of the detection module to select the type of mitigation to be performed. If the interference is frequency modulated, then the IF-based interference mitigation is employed. However, if the interference is classified as periodic interference, e.g., narrow band or CDMA type, then the robust transform domain filtering is employed. If a signal is periodic in the time-domain, e.g., CDMA signal, with period T , then its frequency domain representation becomes sparse with frequency components appearing only at integer multiples of $1/T$, as illustrated in **FIGURE 7**. Such interference can be easily removed by applying non-linearity in the frequency domain.

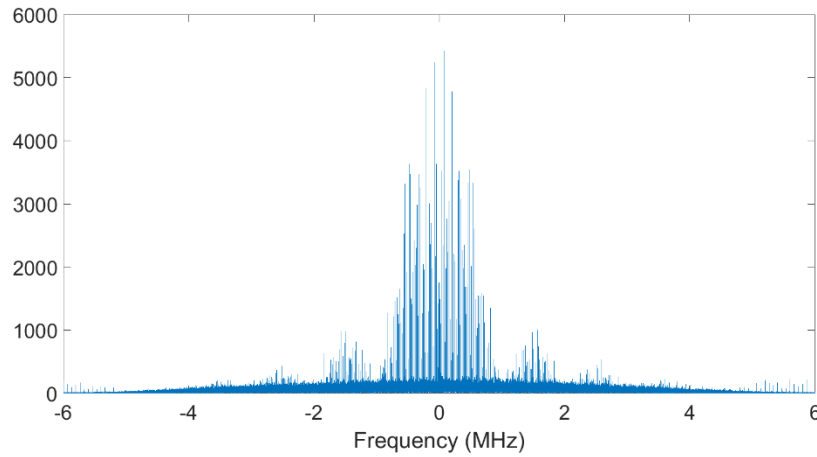


FIGURE 7: Frequency domain representation of CDMA type jamming signal.

The periodic interference mitigation methods are mentioned in the following steps:

- 1) Transform a signal into frequency domain as $X[k] = \sum_n x[n] e^{\frac{j2\pi kn}{N_F}}$, where N_F is the number of samples in the signal. To effectively mitigate the interfering signal, N_F needs to have a higher value as it provides higher frequency resolution to identify bins corresponding to interfering signal.
- 2) Apply Huber's non-linearity:

$$\hat{X}[k] = X[k] \quad \text{if } |X[k]| < T_h; \quad \hat{X}[k] = T_h \frac{X[k]}{|X[k]|} \quad \text{otherwise (eq. 6)}$$

- 3) Inverse transform the signal into time domain as: $x[n] = \left(\frac{1}{N}\right) \sum_k \hat{X}[k] e^{\frac{j2\pi kn}{N_F}}$ (eq. 7)

T_h is a data dependent threshold that is selected so that most of the signal energy is passed undistorted and only bins corresponding to the interference are saturated. We know that most of the desired signal energy is less than $\mu + 2\sigma$ where μ is mean of $|X[k]|$ and σ is its standard deviation. The strong interference appears as spikes in frequency domain that may bias the estimate of μ and σ , so

we use the median in place of mean and median absolute deviation in place of σ . So, the threshold is computed as: $T = \text{median}(|X[k]|) + 2\text{median}(|X[k]| - \text{median}(|X[k]|))$.

Frequency modulated interference mitigation is as follows. If frequency modulated interference is detected, then zero-phase notched filter is applied as [4]:

- 1) The signal is de-chirped as $x_c[n] = x[n]e^{-j\sum_{k=0}^n f[k]}$.
- 2) The notch filter is applied as: $g[n] = \beta g[n-1] + x_c[n] - x_c[n-1]$
- 3) The output of the notched filter is flipped: $\hat{x}[n] = g[-n]$
- 4) The notch filter is applied again: $\hat{y}[n] = k_\alpha \hat{y}[n-1] + \hat{x}[n] - \hat{x}[n-1]$.
- 5) The output is flipped $y_c[n] = \hat{y}[-n]$
- 6) The filtered signal is then chirped again: $y[n] = y_c[n]e^{j\sum_{k=0}^n f[k]}$.

Note that unlike the conventional ANF that performs mitigation online on sample-by-sample basis this method processes stored buffer of sample in both directions. This forward and backward filtering overcomes the non-linear phase distortions common in the conventional ANF.

SPOOFING DETECTION AND MITIGATION

Spoofing detection

In this work, the method of spoofing detection and mitigation focuses on overlapping or aligned spoofing attack where spoofers code delay is within ± 1 chip with an authentic signal. This aligned spoofing will significantly degrade the correlator shape of authentic signals and its local pseudorange (PRN) code [6]. **FIGURE 8** shows examples of correlator shape of an authentic and spoofed GPS L1 C/A signal. When there are no spoofers, the correlator shape of the signal will be a triangle shape (at epoch 40s). However, the correlator shape significantly distorted when an aligned spoofer presents (at epoch 165s). The total of correlator taps for spoofing detection and mitigation is selected to be 15-tap considering the correlation resolution for spoofer estimation as well as computational load to satisfy real-time processing. The 15-tap correlators consist of seven taps for both late and early chips, and one tap for the prompt chip. Ideally, the correlator taps resolutions should be as high as possible to accurately estimate spoofer code delay for spoofing detection and mitigation. **TABLE 1** shows the 15-tap selected to calculate correlation between GPS signal and its local PRN code.

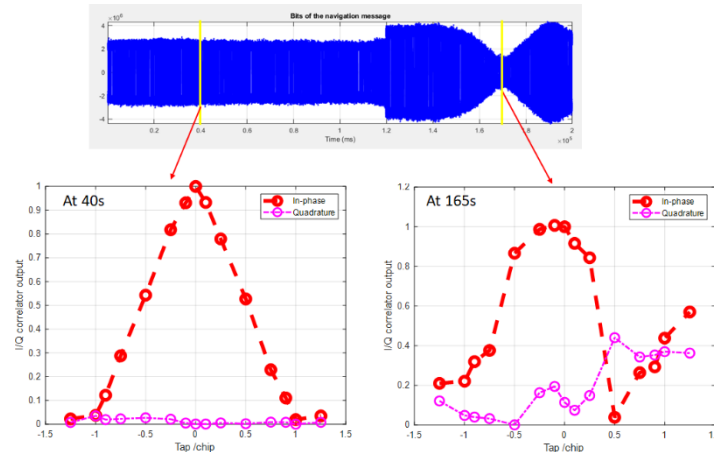


FIGURE 8: Example of correlator shape of GPS L1 signal without (left) and with spoofer (right).

TABLE 1: The list of the correlator taps.

E7	E6	E5	E4	E3	E2	E1	P	L1	L2	L3	L4	L5	L6	L7
-1.25	-1	-0.9	-0.75	-0.5	-0.25	-0.1	0	0.1	0.25	0.5	0.75	0.9	1	1.25

Spoofing detection is performed using a type of deep neural network (DNN) model [7], which was selected following a trade-off analysis among support vector machines (SVM) and convolutional neural networks (CNNs). The number of parameters of the three evaluated models are 53, 8418 and 211437 parameters for SVM, DNN and CNN, respectively. The SVM model has the smallest model size that cause limitations on spoofing detection accuracy [8]. Meanwhile, the CNN, although capable of automatic feature extraction, has a large number of parameters, making training difficult [7]. The DNN model architecture is shown in **FIGURE 9**.

The DNN architecture is a fully connected feed forward neural network with three-hidden layers with tanh activation functions. The selection of this architecture is to trade-off between detection accuracy and detection speed. These models have been trained and tested with TEXTBAT spoofing dataset [9]. **TABLE 2** presents the details of the dataset for training and testing the three ML models. From this dataset, there are a total of 10 signals covering clean and spoofed GPS L1 C/A signal.

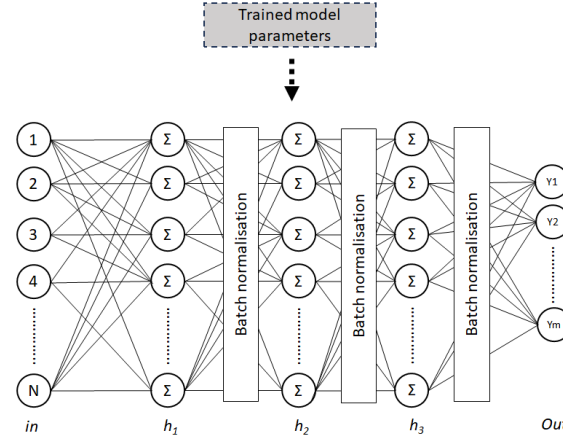


FIGURE 9: The DNN architecture with three hidden layers.

TABLE 2: Detailed TEXTBAT dataset used to train and test the ML models.

DATASET	Description	Total data	Total data points
TEXTBAT	Sampling rate: 25MHz, duration: 550-600s per data set, format: int16 per sample	10 dataset GPS L1= 2 clean datasets (static & dynamic) + 8 spoofed datasets	±1.4 millions of data points. ±40% for authentic signal, ±30% for overlapping at prompt spoofing and ±30% for overlapping at NON-prompt.

The inputs to the ML models are 50 calculated features extracted from the correlator outputs of both of GNSS signals. These features include the ratio of partial data bits [10], slope-based ratios [11], simple ratios, sum ratios, difference ratios [12], residuals of correlation functions [6] and the skewness of correlator points. **TABLE 3** details the description of the calculated features as inputs for the ML models. **FIGURE 10** gives examples of the sensitivity of the features calculated on authentic signal and spoofed signals on selected TEXTBAT signal dataset. From **FIGURE 10**, the calculated features can capture the degradation of correlator shapes calculated when overlapping or aligned spoofers exist.

TABLE 3: Detailed descriptions of the features for ML inputs.

Features	Description
The ratio of partial data bits (beg_end_ratio)	Only on channel I (for DNN and SVM) and for both I and Q channel (for ConvNN)
Slope-based features (MD1, MD2, MD3, MD4)	At four different taps for both I and Q channel
Six Simple ratio	At six different taps (±0.1, ±0.25, ±0.5, ±0.75, ±0.9, ±1) for both I and Q channel
Six Sum ratio	At six different taps (±0.1, ±0.25, ±0.5, ±0.75, ±0.9, ±1) for both I and Q channel
Six Difference ratio	At six different taps (±0.1, ±0.25, ±0.5, ±0.75, ±0.9, ±1) for both I and Q channel
Residual of correlation functions	For both I and Q channel
Skewness of the correlator point	Within a coherent integration cycle (one tracking epoch) for both I and Q channel

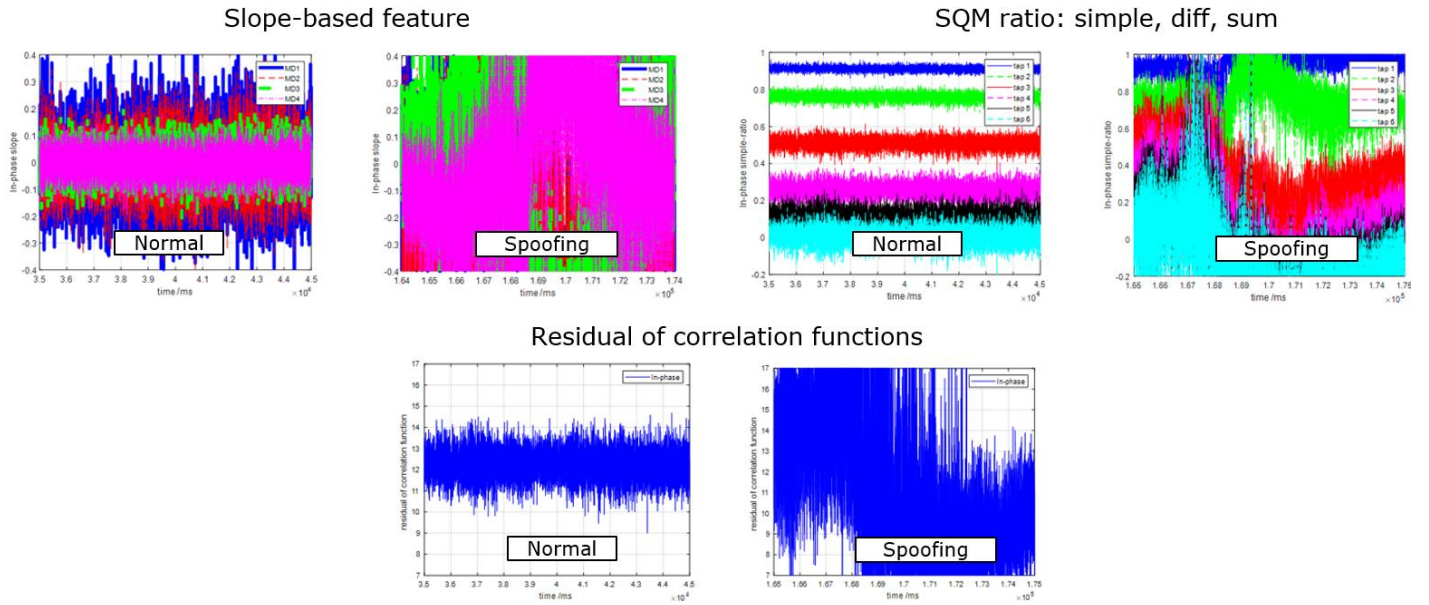


FIGURE 10: Examples of the features sensitivities of correlator outputs on authentic and spoofed signals.

Spoofing detection is classified into 2-class and 3-class detection. **FIGURE 11** shows the 3-class classification where a correlator output is detected as clean (class 1), overlapping at prompt tap (class 2) and overlapping at non-prompt tap (class 3). Meanwhile, 2-class classification only differentiates between clean (class 1) and spoofed situation (class 2+ class 3). Results of spoofing detection are presented in **TABLE 4**. The results show that 3-class classification has lower detection accuracy than 2-class classification. From the results, the DNN model performance, with 3-class training and test accuracy of 99.4% and 98.8% respectively, is the best compared to the SVM and CNN model performance. The CNN, especially for 3-class classification, has the works performance of training and testing accuracy of only 67.4% and 67.7%, respectively that may be caused by insufficient training data to train CNN large model capacity. Note that the data for overlapping at prompt tap is much smaller compared to other class: clean and overlapping at non-prompt tap.

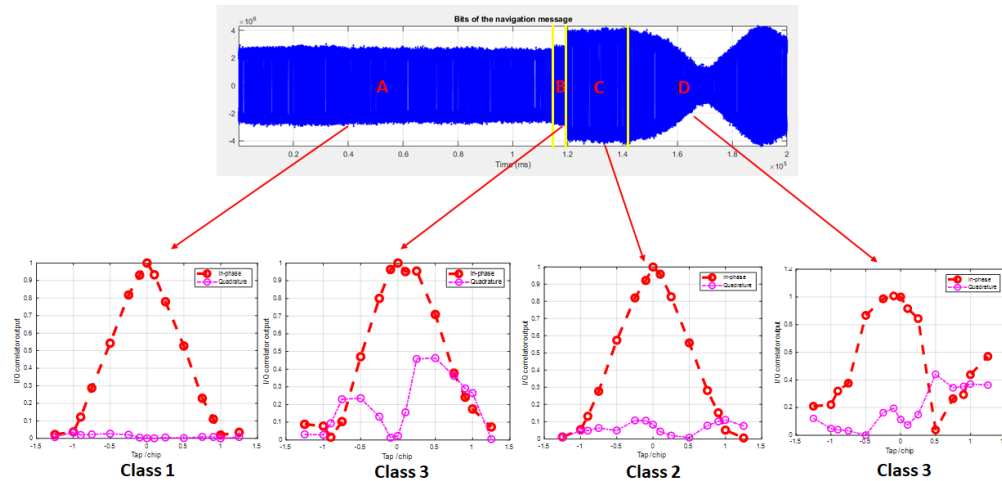


FIGURE 11: 3-class spoofing detection visualization. Note that, for 2-class classification, class 2 (overlapping spoofing at prompt) and class 3 (overlapping spoofing at non-prompt) are combined into a single class.

TABLE 4: Spoofing detection accuracy
for the SVM, DNN and CNN models.

ML types	SVM (Polynomial kernel)		DNN deep neural network (tanh activation function)		CNN deep neural network (ReLU and sigmoid activation function)	
Number of model parameters	53		8418		211437	
Classification type	2-class	3-class	2-class	3-class	2-class	3-class
Training accuracy /%	99.6	98	99.8	99.4	97.6	67.4
Testing accuracy /%	99	97	99.2	98.8	97	67.7

A total of 200s durations of the clean signal of the TEXBAT data set is used to assess the false alarm rate, since the spoofing detection is performed per tracking epoch with 1ms coherent integration time. There is a total of 200000 data points for the false alarm assessment. The spoofing alarm is raised when there are 30 consecutives spoofing detection, that is when there is a 10ms duration of spoofing detection, then a spoofing alarm rises. The false alarm rate of the spoofing detection is 0.009 for signal considered as open-sky signal with no or minimal multipath effect. Meanwhile, for signal with heavy multipath impairments, such as in dense urban areas, the false alarm rate degraded to 0.015. The reason is that overlapping spoofing phenomenon is like multipath phenomenon in terms of degradations of correlator output shapes.

Spoofing mitigation

The spoofing mitigation method focuses on recovering an authentic GPS L1 C/A signal from an aligned match-power spoofing attack, instead of cancelling the signal. The spoofing mitigation algorithm uses an impulse method to calculate the second gradient of correlator outputs of GNSS signals to estimate the code delay of authentic and spoofer signals based on the two largest gradient impulses. The final estimated second-gradient impulse is calculated as a weighted average of three neighbor impulses. In addition, using these impulses, the spoofer's power and phase are estimated. Subsequently, a negative replica of the spoofer signal is generated and subtracted from the original signal (containing both the spoofer and authentic signals). This subtraction process effectively removes the spoofer signal, isolating the authentic signal component.

The impulse method is explained as follows. The impulse is the second gradient of a correlator outputs. The first G^1 and second gradient G^2 (the impulse) are calculated as follows:

$$G^1 = \frac{\Delta corr}{\Delta chip}; G^2 = \frac{\Delta G^1}{\Delta chip^1} \quad (\text{eq. 8})$$

Where $\Delta corr$ is the difference between two consecutives correlator outputs and $\Delta chip$ is the difference between the two consecutive chips. ΔG^1 is the difference between two consecutives first gradient G^1 outputs and $\Delta chip^1$ is the difference between the two consecutive chips after the G^1 calculation.

The accuracy of spoofer parameters estimation, especially the spoofer code delay, highly depends on the chip tap resolutions. Small chip tap distance will have high accuracy estimation. However, the smaller the chip tap distance (high accuracy detection), the higher the sampling frequency required. There is a trade-off between high accuracy spoofer parameter estimations and computational requirements to process the sampled signal. Since the numbers of 15-tap correlators are considered sparse, we need to improve and smoothen the code-delay estimation by using a weighted-average method. The final estimated code delay (tap location) is:

$$\text{Code delay (Tap location) estimate}_i = \frac{(I_{i-1} \times tap_{i-1} + I_i \times tap_i + I_{i+1} \times tap_{i+1})}{(I_{i-1} + I_i + I_{i+1})} \quad (\text{eq. 9})$$

Where I_i is the 2nd gradient impulse at the i -th tap index and Tap_i is the i -th tap index. In short, the refined estimate of the code delay or tap location is based on the neighbor's 2nd gradient impulse at the tap location to be refined. The spoofer is identified as the largest impulse from the two largest gradient impulses.

From these two decomposed signals, we then estimate the power and the IQ carrier phase rotation of the spoofer. The power and IQ rotation θ estimation are as follows:

$$\text{Power}_i = \left(\frac{I_p}{\text{length}(\text{signal})} \right) \times \frac{\text{impulse}_i}{(\text{impulse}_i + \text{impulse}_j)} \quad (\text{eq. 10})$$

$$\theta = \tan^{-1} \left(\frac{Q_p}{I_p} \right) \quad (\text{eq. 11})$$

Where I_p and Q_p are the I and Q correlator output at the p -th tap location (code delay) estimated by the weighted-average of the 2nd gradient impulse. $Power_{adjustment}$ is the adjusted power for each decomposed signal hat is the ratio of its 2nd gradient impulse over the total summation of the two highest 2nd gradient impulse ($impulse_i, impulse_j$).

FIGURE 12: presents the impulse from 15-tap correlator outputs calculated from a nominal GPS L1 signal. In this figure, in nominal condition, the highest impulse (2nd gradient) will be at the prompt tap. Simulations of impulse calculation from spoofed GPS signals at various spoofer delays and 60° carrier phase difference between the authentic and spoofer components is shown in **FIGURE 13:**. In this simulation, the spoofer power is set to be $1.5 \times$ the authentic signal's power.

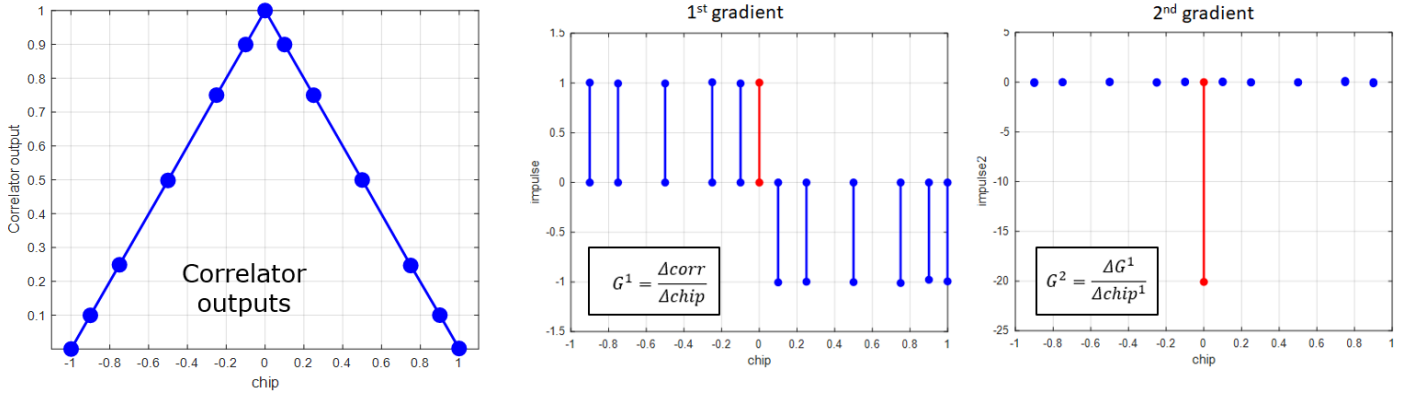


FIGURE 12: Correlator impulse calculated from a nominal GPS L1 signal.

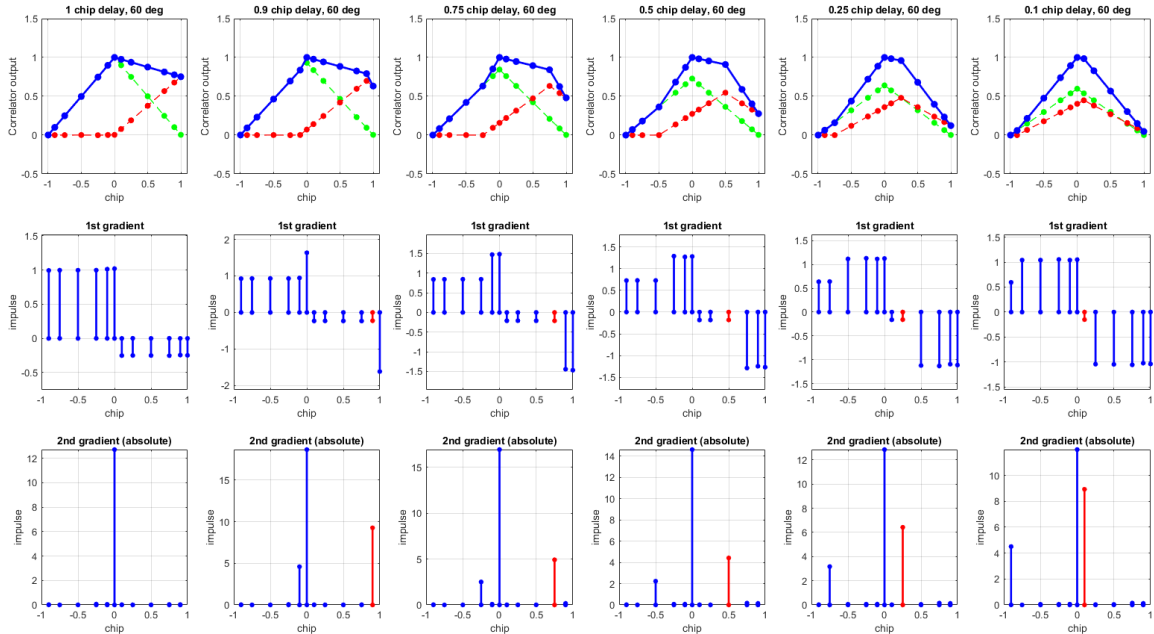


FIGURE 13: Simulations of impulse calculation from spoofed GPS signals at various spoofer delays and 60° carrier phase difference and spoofer power = $1.5 \times$ the authentic signal's power.

FIGURE 14: shows a simulated GPS L1 C/A spoofed signal, containing authentic and spoofer components, and the implementation of the spoofer mitigation (cleaner) processes where the spoofer component (red) is removed leaving only the authentic signal

component (green). The effectiveness of the mitigation depends on how distorted the signal correlator is. To evaluate the mitigation performance, the mitigation method is applied to different tracking epoch (1ms) to TEXTBAT dataset signals at different C/N_0 . The C/N_0 is used as the signal quality measure. **FIGURE 15:** Spoofing mitigation performance under different signal C/N_0 . From performance test results, the spoofing mitigation can be faithfully removed spoofer components and restore authentic components when the C/N_0 is $> 40\text{dB-Hz}$. When the signal has C/N_0 of $< 40\text{dB-Hz}$, the correlator shape distortion is too large, we cannot faithfully estimate the spoofer parameters and remove the spoofer components.

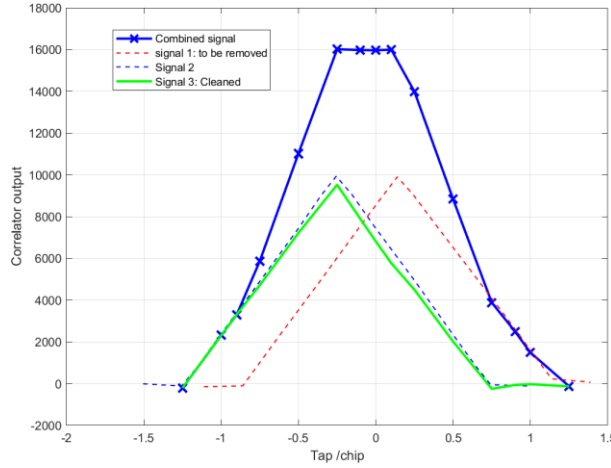


FIGURE 14: An example of spoofer mitigation (cleaner) where the spoofer component (red) is removed leaving only the authentic signal component.

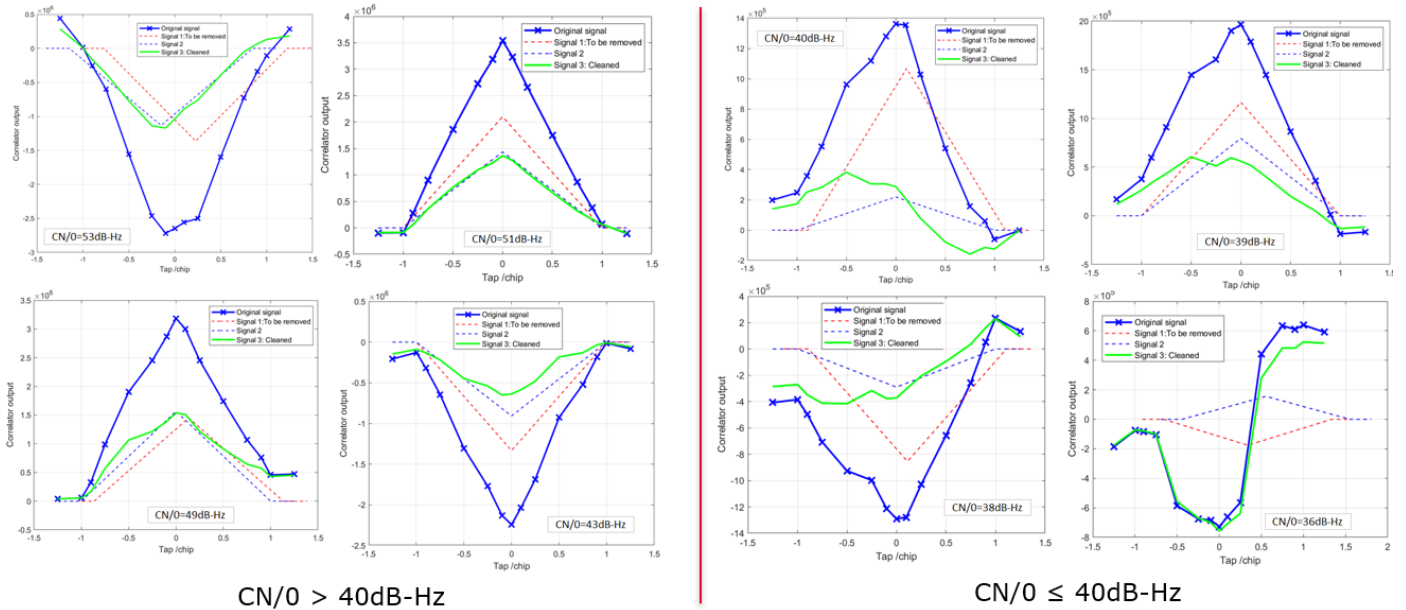


FIGURE 15: Spoofing mitigation performance under different signal C/N_0 (signal quality) from TEXTBAT dataset.

IMPLEMENTATION OF RECEIVER-AGNOSTIC AND REAL-TIME SIGNAL CLEANER

The implementation of the signal cleaner fully utilises real-time software approach under Linux operating system. The main processing unit is a high-performance PC with a multi-core processor. A multi-threading method is utilised to increase computational speed. However, the processing limitation is the interconnection among multi-cores inside the processor that allows only 40% of the total cores (40 cores) of the processor to be utilised. These internal multicore bottlenecks limit the maximum sampling frequency to

operate is 20MHz. **FIGURE 16** shows the schematic view of the developed signal cleaner system. In this figure, there are two receivers used: “system” receiver and “reference” receiver. “system” receiver is the receiver that receive signal after being processed by the system and “reference” signal receives signal directly from the GNSS antenna used for comparison. In **FIGURE 16**, a circular buffer system is used to continuously receive (RF), process (IQ signal) and re-transmit (RF). We use three memory pointers to locate received, processed and transmitted data. In this buffering system, the processing time should be faster than the transmission time to make sure transmitted signal to a downstream receiver is continuous.

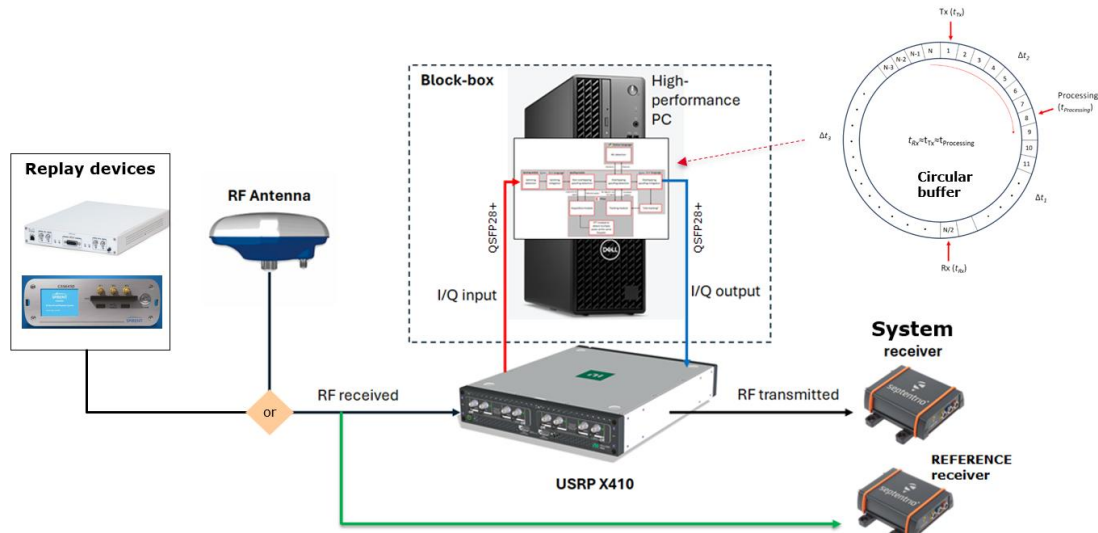


FIGURE 16: The schematic view of the receiver agnostic and real time signal cleaner system.

In summary, the final system building blocks consists of a GNSS RF antenna to capture RF GNSS signals (NavXperience 3G+C Reference with 49dB gain with power input < 50mA), RF front-end (RFFE) based on SDR USRP X410 for RF capture and RF reply of GNSS signals, a processing, including storage, unit to process GNSS signals and implement jamming and/or spoofing detections and mitigations (Intel® Core™ i7 20-Core Processor i7 and 64GB RAM) and a GNSS COTS receiver (Septentrio AsteRx SB3 Pro+) as the downstream receiver to process cleaned signal and provide PVT solutions.

FIGURE 17 Shows the real setup of the system. One of the important components of the system is the RFFE device. After characterizing the RFFE device, the performance of the RFFE device are as follows. The receiver noise is quantified to be -144.8dBm/Hz. The signal dynamic range of the RFFE is 65-75dB-Hz. Meanwhile, the input power dynamic range (where GNSS signal can be faithfully received) of the RFFE is around 85dBm = -50dBm to 135dBm. The transmitter power output is around 17dBm (at 50dB gain) and, by assuming linearity, the power output is -23dBm (at 10dB gain) and at -33dBm (at 0 dB gain).

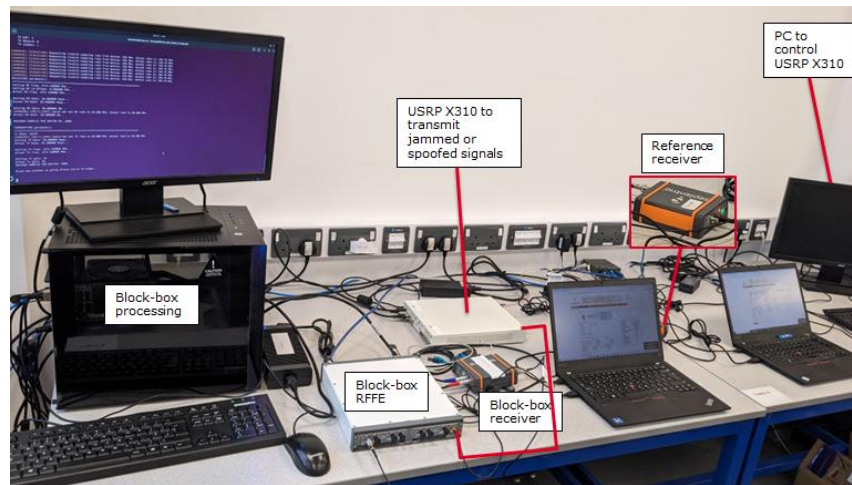


FIGURE 17: The real setup of the developed signal cleaner system.

EXPERIMENT RESULTS

The system is tested through experimental campaigns using real datasets obtained from The Jammer Test 2024 in Andøya, Norway (for jamming mitigation) and TEXBAT dataset (for spoofing mitigation) by using a professional commercial GNSS receiver. The Jammer test dataset is obtained from ESA and covers various types of jammers, including a low-powered chirp jammer, a ramp-up-and-down CDMA jammer, and a narrow-band jammer with a slow-varying center frequency.

Low Powered Jammer Attack

A low powered pulsed jammer (0.2W) with clean first 90 seconds is replayed using Spirent transmitter to both reference and developed system receiver. Reference receiver performance was tested in two modes: a) internal mitigation on and b) internal mitigation off. **FIGURE 18** (left) illustrates the signal detection performance as well as Spectrogram of jammed signal.

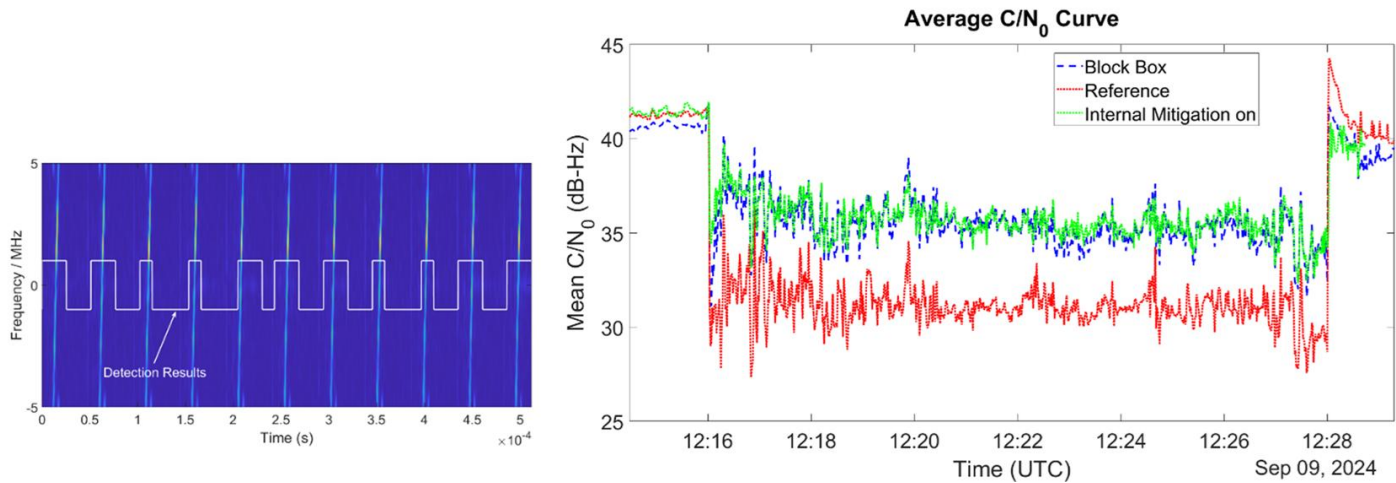


FIGURE 18: (left) Spectrogram of the jammer and signal detection and (right) Impact of interference mitigation using instantaneous position error and C/N_0 .

The interference observed in the given signal is pulsed interference. To evaluate the interference characterization performance of the system, we need to compare the output of the system with the ground truth. The interference appears only for short duration and affects only a few samples, so we obtain the ground truth by dividing the signal into segments of 128 samples computing energy for each segment. If energy exceeds manually adjusted threshold, interference is said to be detected in the segment otherwise not. The probability of detection by comparing the developed system output with energy-based detector is 0.98 and probability of false alarm is 0.08 if the complete signal is used for analysis. However, if we compute the probability of false alarm of interference the free part of signal is 0.04. The performance comparison between the system and reference receiver is performed using average C/N_0 is illustrated in **FIGURE 18** (right). The interference mitigation results in approximate 5 dB-Hz gain in C/N_0 . Results show that internal mitigation offers 0.25 dB advantage in terms of C/N_0 as compared to block-box during interference part of signal. The block-box system itself introduces approximately 1.5 dB distortion in interference free part so block-box algorithm will give approximately 1.25 dB gain if the block-box algorithm is implemented within GNSS receiver. **TABLE 5** summaries the impact of interference mitigation on low powered jammer in terms of C/N_0 .

TABLE 5: Impact of interference mitigation on low powered jammer in terms of C/N_0 .

	Block-box/dB-Hz	Without Internal Mitigation/dB-Hz	With Internal Mitigation/dB-Hz
Interference Free C/N_0	40.15	41.39	41.7
With Interference C/N_0	35.3	31.2	35.54

Ramp-up Ramp-down jammer attack

A CDMA type ramp-up ramp-down jammer superimposed on GNSS signals was replayed through Spirent. The jammer power is increased from -37 dBm with step size of 2dBm to 47 dBm and back. The interference mitigation is performed in real-time using the

developed system. The impact of interference mitigation in terms of extraction of position error and average C/N_0 is illustrated in **FIGURE 19**.

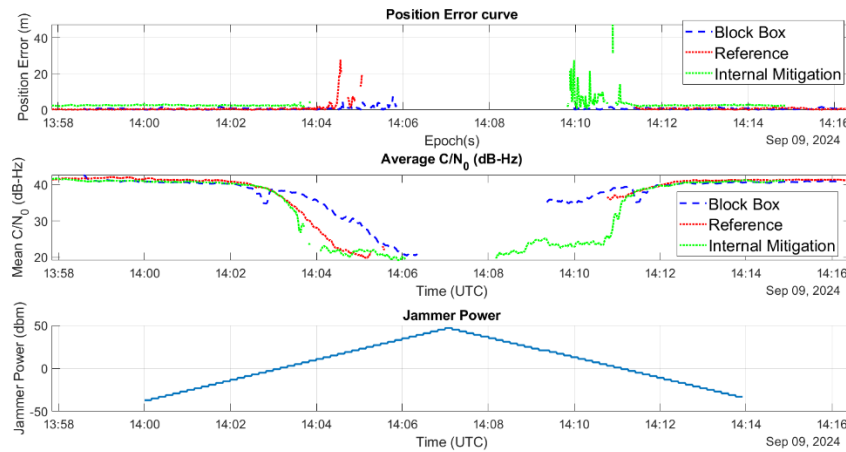


FIGURE 19: Impact of interference mitigation on position solution for CDMA attack.

It is observed that block-box based interference mitigation allows receiver to track up to power levels of 31 dBm and it then restarts tracking at 13 dBm. The reference receiver with no interference mitigation can track up to 19 dBm and regain tracking at -3 dBm. The reference receiver with interference mitigation on state loses track around 15 dBm jammer power level. It restarts tracking at the same time as block-box based receiver but with large position error.

Narrowband Interference

A jammer of power level 50W emitting a continuous wave that drifts from 1545 MHz to 1620 MHz in 15 minutes is added to a GNSS signal of a moving object. The instantaneous frequency of the jamming signal as it appears in the 10 MHz band of the signal is illustrated in **FIGURE 20** (left). The developed system classifies the interference as periodic interference and detects it with the probability of detection equal to 0.99 with false alarm equal to 0.0023. The recorded signal is replayed through Spirent. The impact of interference mitigation in terms of the average C/N_0 is shown in **FIGURE 20** (right).

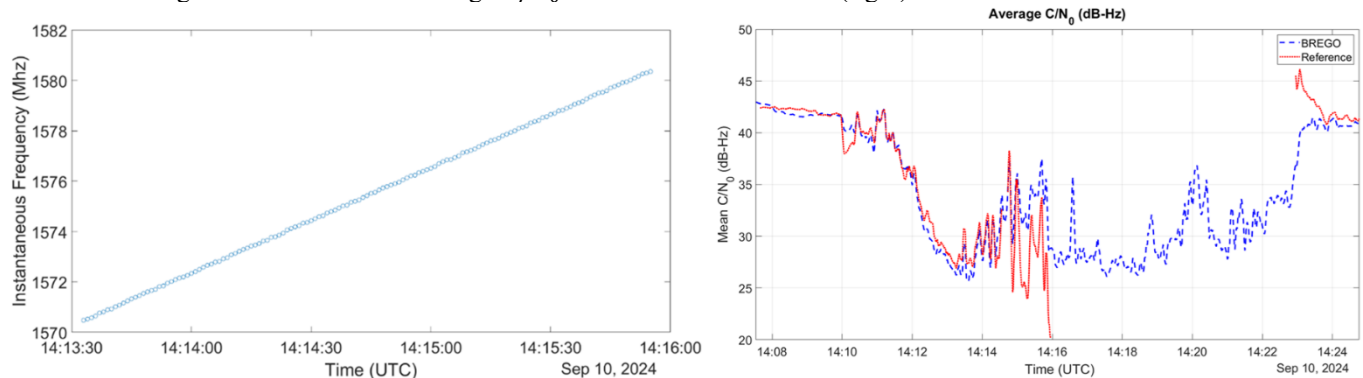


FIGURE 20: (left) Instantaneous frequency of the CW jamming attack as it enters the 10MHz spectrum and (right) Impact of interference mitigation on Continuous wave attack.

It is observed that after 14:16, the block box achieves significant improvement over the reference receiver as the reference receiver fails to estimate C/N_0 . The performance of reference receiver deteriorates as the frequency of interfering signal comes in the centre of L1 band. However, the GNSS receiver can effectively mitigate the interference and extract position curve as illustrated in **FIGURE 21**. Note that without interference mitigation, the complete path cannot be extracted.

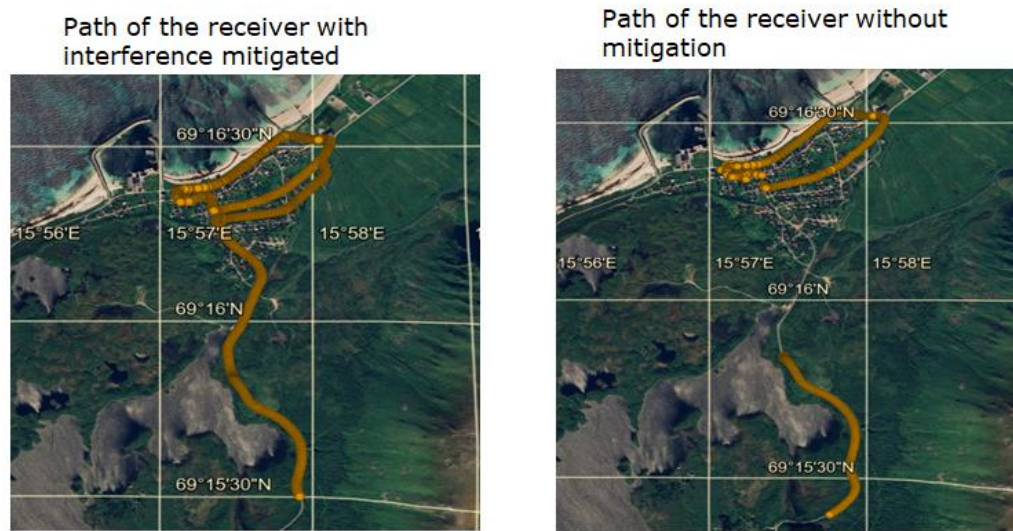


FIGURE 21: Positions for the moving object with and without interference mitigation.

Spoofing TEXBAT ds3 dataset

The spoofing mitigation experiments use the TEXBAT ds3 dataset [9], containing a spoofer with 1.3 dB higher power than the authentic signal. In this data set, a spoofer is injected from 120s and the spoofing starts approximately between 150-200s. This dataset contains spoofed GPS L1 signals, where a static receiver initially tracks an authentic signal before encountering a spoofer, resulting in a significant shift in the static location for approximately 50 seconds. **FIGURE 22** shows the results of the spoofing mitigation by using the impulse method. From these results, the spoofing mitigation shows a reduction in static position error by up to 70% while also improving the duration of position availability.

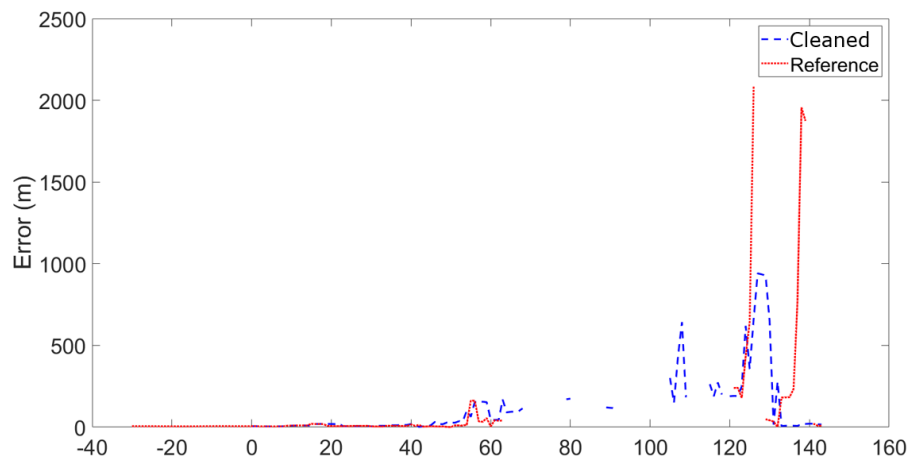


FIGURE 22: Position error between spoofed and cleaned signal.

Results from both jamming and spoofing mitigation evaluations demonstrate the potential benefits of the developed system in providing clean signals to a wide range of receiver types.

CONCLUSION AND FUTURE WORK

In conclusion, we have demonstrated that the developed real-time and receiver-agnostic jamming and spoofing detection and mitigation system is effective on real-world data. Potential applications for the system include securing GNSS stations that require continuous, jamming- and spoofing-free GNSS signal recordings to provide publicly accessible data, as well as safeguarding

receivers for critical assets. The system has been validated using a professional (commercial) GNSS receiver and a software receiver (GNSS-SDR). The system is flexible and configurable and can use different radio frequency front-end and processing unit devices. A novel variant of the developed adaptive notch filter can track fast time-varying chirps because of an additional post-processing step and mitigates interferences without causing non-linear phase distortions by employing zero-phase filtering. The novel spoofer mitigation technique is able to remove the spoofer component from the TEXBAT ds3 signal. The developed system has certain limitations, including restricted computational speed due to CPU constraints, reduced effectiveness when mitigating interference from multiple equally powerful chirp jammers, and a dependence on the receiver's ability to initially track authentic signals, as well as on specific signal structures for successful mitigation.

Future work will focus on testing with more spoofing scenarios to improve the current mitigation algorithms, integrating feedback from receivers and other sources for spoofing detection and mitigation in wider scenarios and implementing jamming and spoofing detection and mitigation on field programmable grid array (FPGA) for fast processing to create a compact, portable system suitable for a wider range of applications including dynamic environments and moving assets.

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REFERENCES

- [1] Misra, P, Enge, P., "*Global Positioning System: Signals, Measurement and Performance*", 2nd Edition, 2006, Ganga-Jamuna Press: USA.
- [2] Kaplan, E.D. and Hegarty, C., "*Understanding GPS/GNSS: principles and applications*", 2017. Artech house.
- [3] Dovis, F., "*GNSS interference threats and countermeasures*", 2015, Artech House.
- [4] Khan, Nabeel Ali, and Luis Enrique Aguado. "Adaptive Notch Filter based Interference Characterization and Mitigation for GNSS Receivers." *Proceedings of the 37th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2024)*. 2024.
- [5] Kormylo, J., & Jain, V., "*Two-pass recursive digital filter with zero phase shift*", 1974 IEEE Transactions on Acoustics, Speech, and Signal Processing, 22(5), 384-387.
- [6] Wesson, K.D., Shepard, D.P., Bhatti, J.A. and Humphreys, T.E., "*An evaluation of the vestigial signal defense for civil GPS anti-spoofing*". In Proceedings of the 24th International Technical Meeting of the Satellite Division of the institute of navigation (ION GNSS 2011) (pp. 2646-2656) 2011, September.
- [7] Goodfellow, I., Bengio, Y., Courville, A. and Bengio, Y., "*Deep learning*", 2016, Cambridge: MIT press.
- [8] Murphy, K.P., "*Machine learning: a probabilistic perspective*", 2012. MIT press.
- [9] Humphreys, T.E., Bhatti, J.A., Shepard, D. and Wesson, K., 2012. *The Texas spoofing test battery: Toward a standard for evaluating GPS signal authentication techniques*.
- [10] Seco-Granados, G., Gómez-Casco, D., López-Salcedo, J.A. and Fernández-Hernández, I., "*Detection of replay attacks to GNSS based on partial correlations and authentication data unpredictability*", 2021, Gps Solutions, 25(2), p.33.
- [11] Khan, A.M., Iqbal, N., Khan, A.A., Khan, M.F. and Ahmad, A., "*Detection of intermediate spoofing attack on global navigation satellite system receiver through slope-based metrics*", 2020, The Journal of Navigation, 73(5), pp.1052-1068.
- [12] Negrinho, A., Boto, P., Fernandes, P., Mendonça, V., Nunes, F. and Sousa, F., "*Signal Quality Monitoring Aspects in GNSS Signals Affected by Evil Waveforms*", In Proceedings of the 35th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2022) (pp. 3947-3958), 2022, September.